Textile ERP Backend: Audit Report

Tareekh: 21 Oct 2026 | Scope: poora backend code, database design, aur live (Supabase) database (sirf parha) | Zubaan: Roman Urdu

1. Natija ek nazar mein

> Core costing theek hai. Lagbhag 1,900 random operations ke baad bhi stock aur paisa ek paisa idhar-udhar nahi hua. Audit mein 17 masle mile (AUD-01 se AUD-17) aur 6 database/config notes. **AUD-04 ke siwa sab theek kar diye gaye** aur har ek ka test ab pass hota hai.

Sawal	Jawab
Costing kharab to nahi?	Core (khareed, average rate, batch issue, return, adjustment, rework) theek. Teen calculation masle mile (AUD-01, 02, 13), teeno fix.
Kapre ka hisaab?	Har meter grey store se gate pass tak gina jata hai. Kapra na banta hai na gum hota hai.
Loopholes?	Processed job reject, packing mein zyada meters, gate pass ke baad delete. Sab band kiye.
Security?	Company isolation: 3,300+ hamle, sab refuse. Tokens, roles, VIEW-only theek.
Crash (500)?	3 qism (ghalat tareekh, purana accept, bade numbers). Fix.
Live database?	37 accounting rules mein se 37 theek.
Speed?	1,000 lots/cards/jobs par har screen 0.5 s se kam, N+1 nahi.

2. Tareeqa

- **37 invariants** (SQL rules): jaise "item ka stock = batches ka jor", "lot outward = SO cards + wapas", "packing list = Folding ka bheja hua". Har test ke baad aur live DB par chalaye.
- **Shadow model:** random operations ko alag FIFO model se muqabla (12 seeds x 160 qadam).
- **Scenarios:** costing 12, kapra 21, mushkil flows 3, concurrency 12, reports ka mel 5, security 8 (1,803 GET/DELETE calls + 60+ valid hamle), kharab input (8,986 calls), super admin (1,485 calls), audit log (54 calls), time zone, extremes, Urdu/emoji, speed.
- **Total:** 36 suites, 182 tests: 181 pass, 1 skip (speed), 0 fail. Aik `test.failing` baqi (AUD-04).

3. Masle aur unki halat

HIGH

ID	Masla	Asar	Fix
AUD-05	Packing list line ki meters par koi hadd nahi (100 m thaan par 5000 m).	Invoice par mojood na kapra aur ghalat amount.	Fixed line <= Folding ka bheja (thaan + grade, saari lists mila kar)
AUD-06	Gate pass ke baad list reopen + delete; gate pass bhi cascade se mit jata.	Dispatch ka record ghayab.	Fixed `HAS_GATE_PASS`
AUD-03	COMPLETE -> PENDING -> REJECTED; chemical lag chuke the.	Cloth ke cost-raaste mein Rs 0, store mein Rs 1000.	Fixed chemical request wala job reject nahi

AUD-01	Folding kapra 2 hisson mein bheje to Packing screens par cost doguna (Rs 2000 vs 1000).	Sirf Packing ke cost columns ghalat.	Fixed `previousCosts` meters ke hisse se
AUD-12	SO card status/department/deliveryText pehle department par atka.	Ghalat location; report ka "Last sent to".	Fixed position jobs se

MEDIUM

ID	Masla	Fix
AUD-04	Folding grading par koi hadd nahi (200 m aya, 5000 m grade).	Open business faisla (gain/loss tolerance) baqi
AUD-07	Ghalat tareekh (2026-02-30) par 500, body aur ~25 filters.	Fixed asli date check, 400
AUD-08	Thaan wali lot plain total se edit ho jati thi.	Fixed `LOT_HAS_THAANS`
AUD-09	Employee kisi bhi SO card ka raasta dekh sakta tha.	Fixed department check, 403
AUD-11	Purana `/so-cards/:id/accept` 500.	Fixed 422 `ACCEPT_ON_JOB`
AUD-13	Packing total aur lines ka jor 2 paisa alag (48.35 vs 48.37).	Fixed total = lines ka jor
AUD-14	Default tareekh/day filters UTC, Pakistan nahi (11 jagah).	Fixed Asia/Karachi
AUD-15	Store request se zyada chemical bhej sakta tha.	Fixed `MORE_THAN_REQUESTED`

LOW aur database/config

ID	Masla	Fix
AUD-02	Half-paisa rounding neeche (har ~1,470 lines mein 1 paisa kam).	Fixed `src/utils/money.js`
AUD-16	qty x rate bohot bada: 500.	Fixed 400 `NUMBER_TOO_BIG`
AUD-17	Tareekh ki sanity nahi (card lot se pehle).	Fixed
AUD-18	81 foreign keys par index nahi.	Fixed migration `20261021000001`
AUD-19	39 numeric columns par CHECK nahi.	Fixed 25 columns, `CHECK >= 0 NOT VALID`
AUD-20	Folding list/send-form poora data uthate hain.	Not changed 1,000 jobs tak theek
AUD-21	Dev `.env`: chhote JWT secrets, 1 minute lock, `trust proxy`.	Not changed deployment setting
AUD-22	Purana data (11 jobs bina received stamp, ~25 bina operator, 2 purani packing lists).	Not changed nuqsan nahi

AUD-23	Recursive queries mein cycle guard nahi; `for(;;)` code generator.	Fixed `UNION`, 50 koshish

4. Jo cheezein theek nikli

- Stock/value hamesha batches ke jor ke barabar; negative kabhi nahi; ledger match.
- Average rate aur batch costing: 20 kg @40 + 100 kg @45, 30 kg issue = Rs 1250 (800+450).
- Return usi rate par; rework loop mein cost sirf ek baar; cost bantwara 5 jagahon par ek jaisa.
- Concurrency: 8 SO cards ek lot se ek waqt mein, bilkul 3 kamyab; numbering mein duplicate nahi.
- Passwords bcrypt 12 rounds; `npm audit` 0 vulnerabilities; git mein secrets nahi; SQL injection safe.
- Audit log immutable; 40 migrations ulti/seedhi chalti hain.

5. Baqi kaam aur faisle

1. **AUD-04:** Folding mein kitna gain/loss theek hai (jaise 5%)? Us se zyada par rok ya approval?
2. Migration `20261021000001` live DB par abhi nahi chali: `npm run db:migrate`, backup ke saath.
3. Production `.env`: JWT secrets 32+ characters, login lock 15 minute, nginx ke peeche.
4. Frontend ko 5 jagah react karna hai (guide: `docs/frontend/audit-fixes.md`).

6. Dobara chalane ka tareeqa

`node scripts/audit-db.js`	Live DB ke 37 rules (read-only)
`npx jest --runInBand`	Saare tests (UTF8 `erp_test` database chahiye)
`AUDIT_PERF=1 npx jest tests/audit/performance.test.js`	Speed test, lagbhag 4 minute

7. Hadden

Tests local Postgres (UTF8) par chale, Supabase pooler par nahi; live DB par sirf SELECT. Speed 1,000 jobs tak naapi. Frontend, browser, nginx, backup aur production `.env` dekhe nahi gaye. Business rules aap ke faislon par hain.